

Machine Learning with Graphs: Surveys

Arlei Silva

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Students enrolled in COMP 559 will write and review surveys; those enrolled in COMP 459 will only review. Surveys should be written in groups of up to 3 students. Each group will read at least 10 related papers on a topic that is relevant but not covered in the course and write a short survey. The survey should describe the general problem, some of the existing approaches for the problem, and how they compare with each other, and possible new research directions. These are some examples of topics:

- Graph sparsification;
- Graphons;
- High-performance graph machine learning;
- Reinforcement learning on graphs;
- Games on social networks.

Keep in mind that the goal of the survey is not simply to summarize each paper, but also to provide insights about the topic by leveraging ideas from the papers and your own insights.

Survey proposal (02/10)

Each group should propose a topic and a list of at least 10 papers.

Survey (4/24)

The length of final reports is limited to six pages using the NeurIPS format.¹ Reports are expected to be original, clear, grammatically correct, well-organized, and technically sound.

¹<https://nips.cc/Conferences/2020/PaperInformation/StyleFiles>

Survey reviews (4/29)

All students will be asked to review a subset of at most three surveys. Each review will be a one-page document (also NeurIPS format) with a paragraph summarizing the survey, followed by an evaluation of the survey in terms of (1) clarity and organization, (2) relevance to the course, and (3) coverage of the area. Each evaluation criterion can be described in a short paragraph and evaluated as either *needs improvement*, *good*, or *very good*. The entire survey should also be rated as *needs improvement*, *good*, or *very good*. Reviewers are not expected to read the specific papers covered in the surveys.